Normal distribution + Z standardisation

A1S2 PRE-ASSESSMENT STUDY

1 RECOGNISE IT

- Bell-shaped, symmetric data with mean μ and SD σ.
- Course model notation is $X \sim N(\mu, \sigma^2)$.
- Need to move between original X units and standard Z units.

2 DECISION / METHOD

1. Verify X is modelled as $X \sim N(\mu, \sigma^2)$.
2. Standardise x to z using σ.
3. Use the sign as a reasonableness check.
4. Back-transform only if the answer must be in X units.

3 FORMULA

$X \sim N(\mu, \sigma^2)$

$z = (x - \mu)/\sigma$

$x = \mu + z\sigma$

$Z \sim N(0,1)$

4 FORMULA SHEET VS WHAT I KNOW

Gives: the inference table uses Z when population σ is known.
Still know: raw-score standardisation, back-transformation, and that positive z is above the mean.

5 EXCEL ROUTE

- Direct: `=NORM.DIST(x,mu,sigma,TRUE)`
- Standard: `=NORM.S.DIST(z,TRUE)`
- Do not enter variance where Excel asks for SD.

6 TINY WORKED EXAMPLE

If μ=72, σ=8 and x=84, then z=(84-72)/8=1.5: x is 1.5 SD above the mean.

7 TOP EXAM TRAPS

- COURSE NOTATION uses σ^2 as the second Normal parameter.
- EXCEL NORM.DIST asks for σ, not σ^2.
- Using SE=σ/√n for one observation X.
- Losing the sign of z.

8 SAY ALOUD

Z tells me how many standard deviations the observation is from the population mean.

9 EXIT CHECK

For x below μ, what sign must z have, and why?

ONE-GLANCE VISUAL ROUTE

